LIANGSHAN ENTERPRISE

THREAT ANALYSIS

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Submitted to: M. Abdullah Abid

ii. Threat Analysis

Threat 01

Unauthorized access could lead to data breaches involving architectural designs, client data, or proprietary research. **(Unauthorized Access to Sensitive Data)**

Severity: High

Mitigation: Already mitigated through

- Role-Based Access Control (RBAC) for sensitive data.
- AES-256 encryption for data at rest and in transit.
- Regular audits of access logs to prevent unauthorized access.

Threat 02

Malicious or negligent insiders accessing or manipulating sensitive data. **(Insider Threat)**

Severity: High

Mitigation: Already mitigated through:

- Role-Based Access Control (RBAC) limiting employee access based on roles.
- Real-time anomaly detection systems leveraging AI to identify unusual patterns.
- Employee training programs to instill a culture of data accountability and security awareness.

Threat 03

Employees might fall victim to phishing emails, leading to credential theft or unauthorized system access. **(Human Error/Failure)**

Severity: Medium

Mitigation: Not mitigated, solutions are:

- Regular cybersecurity training for employees.
- Deployment of email filtering tools to block phishing attempts.
- Multi-factor authentication (MFA) to secure employee accounts.

Threat 04

Infection through malicious emails, compromised websites, or external devices can disrupt operations and encrypt critical data. **(Software Attacks)**

Severity: **High**

Mitigation: Already mitigated through:

- Endpoint protection solutions and antivirus tools.
- Network monitoring with IDS/IPS systems for early detection.
- Incremental and differential data backups replicated across secure locations.

Threat 05

Distributed Denial of Service (DDoS) attacks could target Liangshan's global MPLS network, causing operational disruptions. **(deviation in service)**

Severity: **Medium**

Mitigation: Not mitigated, solutions are:

- Use of cloud-based DDoS protection services.
- Load balancers and redundancy in network design.
- Monitoring and blocking suspicious traffic.

Threat 06

Breaches of physical security at headquarters or regional offices. **(Espionage or Trespass)**

Severity: **High**

Mitigation: Already mitigated through:

- Biometric access controls and RFID-enabled badges.
- 24/7 CCTV surveillance and real-time monitoring.
- Visitor management systems with electronic logging.

Threat 07

Disasters could disrupt operations at headquarters, data centers, or regional offices. **(Forces of Nature)**

Severity: **High**

Mitigation: Already mitigated through:

- Backup sites and disaster recovery plans.
- Data centers with redundant power supplies and advanced cooling systems.
- Disaster-resilient facilities for off-site data backups.

Threat 08

Fire incidents at critical facilities such as data centers could result in data loss or operational downtime.

Severity: **Medium**

Mitigation: Already mitigated through:

- Advanced fire suppression mechanisms integrated into building designs.
- Use of fire-resistant materials.

Threat 09

IT system failures could impact project management, communication, and data availability. **(Technical Hardware & Software Failures/Errors)**

Severity: **High**

Mitigation: Already mitigated through:

- Hybrid cloud solutions with high availability for IT services.
- Incremental and differential backups stored in secure locations.
- Real-time monitoring of system performance using AI tools.

Threat 10

Sudden changes in international or regional regulations could disrupt operations.

Severity: **Medium**

Mitigation: Not mitigated, solutions are:

- Proactive monitoring of regulatory environments.
- Flexible compliance frameworks that can adapt to new requirements.

Threat 11

Exploitation of vulnerabilities in IoT devices used in smart buildings and offices.

Severity: Medium

Mitigation: Not mitigated, solutions are:

- Regular firmware updates for IoT devices.
- Network segmentation to isolate IoT devices.
- AI-based monitoring of IoT devices for anomaly detection.

Threat 12

Failure to comply with GDPR, CCPA, or China's Cybersecurity Law could result in legal penalties and reputational damage.

Severity: High

Mitigation: Already mitigated through:

- By maintaining updated compliance frameworks aligned with regulations.
- Conducting regular audits and staff training on data privacy laws.
- Employing dedicated compliance officers to oversee global operations.